Computer Communication and Networks

Data Link Layer: Introduction, Framing & Control

Session 7

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Data Link Layer

Learning Outcome:- At the end of the lesson, you are expected to be able to:

- Define framing in data link layer
- Identify types of framing protocols
- Explain the bit stuffing and destuffing
- Identify the different controls in data link layer.



Data Link Layer: Introduction

Introduction:

- DLL's specific responsibilities includes framing, addressing, flow control, error control, and media access control.
- DLL divides the stream of bits received from network layer into manageable data units called **frames**.
- DLL imposes the **Flow control mechanism** to avoid overwhelming the receiver.
- DLL also imposes the **Error control mechanism** to detect and retransmit the damaged, duplicated or lost frames.



Data Link Layer: Framing

Framing:

- Framing is the process of dividing the stream of bits into manageable units.
  - Frames can be of **fixed-size** and **variable-size**.
    - In fixed-size frames, no need to define the boundary of frames; the size itself can be used as a delimiter.
    - Variable-size framing is commonly used in LAN
      - ✓ In **variable-size frame**, we need to define end of the frame and beginning of next frame.
- Two approaches were used in **variable-size** framing:
  - ✓ Character-oriented approach
  - ✓ Bit-oriented approach



Data Link Layer: Framing: Character-Oriented Approach

Character-Oriented Protocol:

- It is also known as **Byte-Oriented Protocol**
- In character-oriented protocol:-
  - data to be carried are 8-bits character from the coding system such as ASCII
    - ✓ To represent numeric, alphabetic, and special characters in a computer's internal storage and on magnetic media, we must use some sort of coding system.
    - ✓ In computers, the code is made up of fixed size groups of binary positions.
      - Each binary position in a group is assigned a specific value; for example 8, 4, 2, or 1.
    - ✓ In this way, every character can be represented by a combination of bits that is different from any other combination.

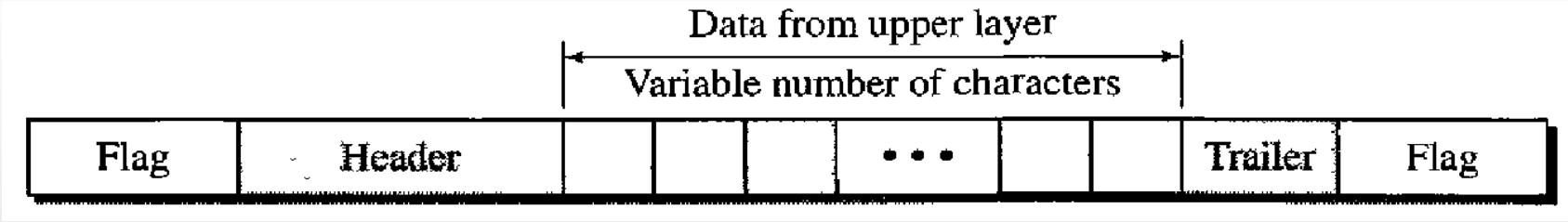
Source: https://www.tpub.com/neets/book22/94a.htm
https://www.geeksforgeeks.org/different-types-of-coding-schemes-to-represent-data/



Data Link Layer: Framing: Character-Oriented Approach

Character-Oriented Protocol:

- It is also known as **Byte-Oriented Protocol**
- In character-oriented protocol:-
  - Header, Trailer and Flag are added.





Data Link Layer: Framing: Character-Oriented Approach

Character-Oriented Protocol:

Header:-

- It contains the source & destination addresses and
- Other control information.

Trailer:-

- It carries error detection redundant bits or
- It carries error correction redundant bits

Flag:-

- To separate one frame from the next, 8-bits(1 byte) **flag** is added at the beginning and end of the frame.



Data Link Layer: Framing: Character-Oriented Approach

Character-Oriented Protocol:

- Character-Oriented Framing:-
  - It is **mostly popular when only text was exchanged by DLL.**

- In Character-Oriented framing, the byte-stuffing (character-stuffing) takes place.

- **Byte-stuffing**:-
  - A special **byte** is added to the data section of the frame when there is a character with the same pattern as a flag.



Data Link Layer: Framing: Character-Oriented Approach

❖ Character-Oriented Protocol: Byte-stuffing

➢ Byte-Stuffing:-

- This byte stuff is an extra byte usually called the **escape character** (ESC), which has a predefined bit pattern.

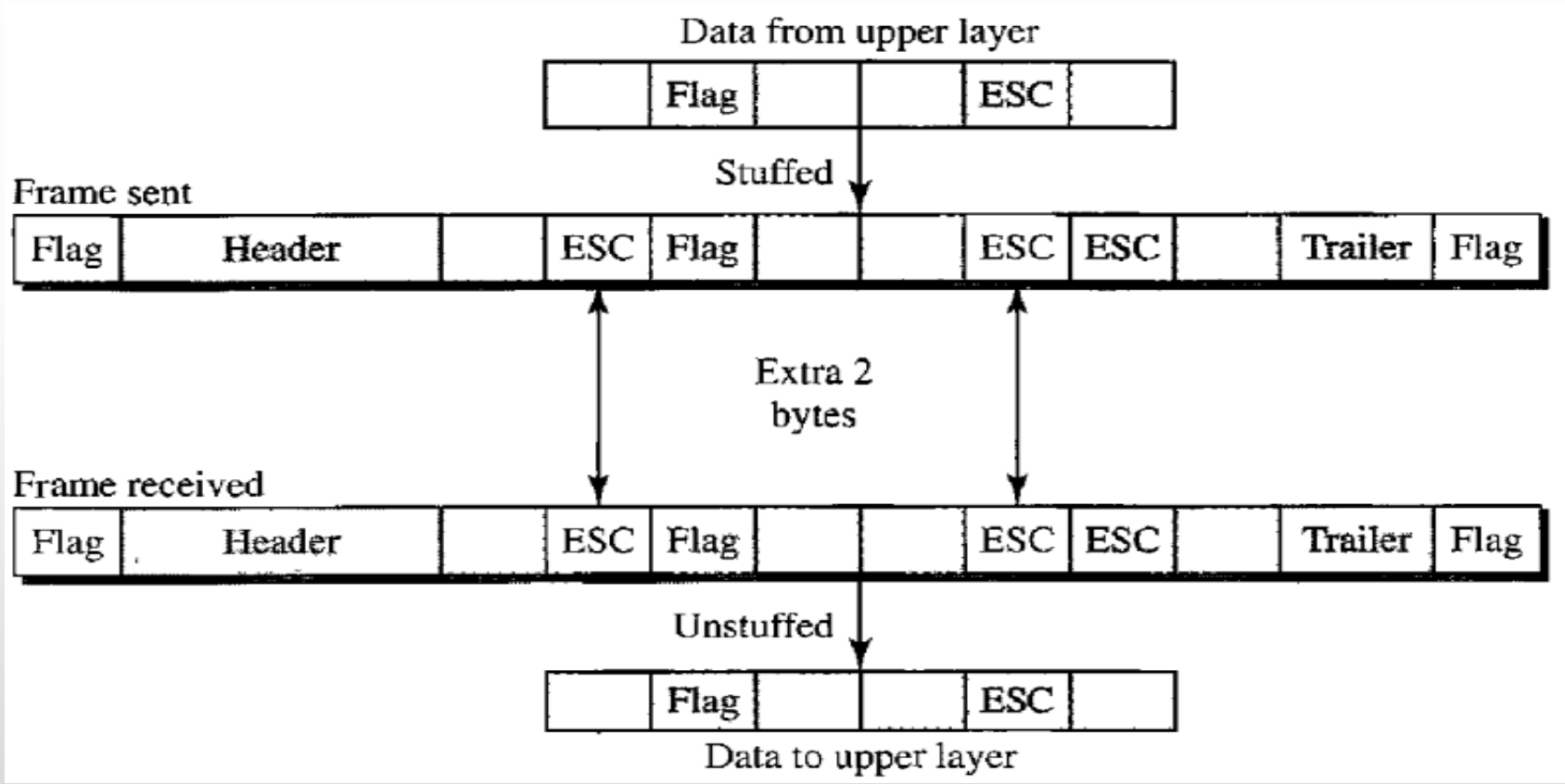
- Whenever the receiver encounters **ESC** character, it removes from the data section and treats the next character as data.



Data Link Layer: Framing: Character-Oriented Approach

❖ Character-Oriented Protocol: Byte-stuffing

- Figure shows the byte stuffing and unstuffing.





Data Link Layer: Framing: Bit-Oriented Approach

❖ Character Oriented Protocol Issue:

- ➢ Character-Oriented gives a problem in **data communications**.
  - Universal coding systems are used today such as Unicode:-
    - Unicode is either 16-bit & 32-bit characters
      - 16-bit character sets a standard, designed to include characters appearing in most languages including Chinese, Japanese, etc.
    - ✓ Therefore, data communications with such systems in place, there is conflicts with 8-bit characters.
  - Therefore, tendency is moving to bit-oriented protocol.



Data Link Layer: Framing: Bit-Oriented Approach

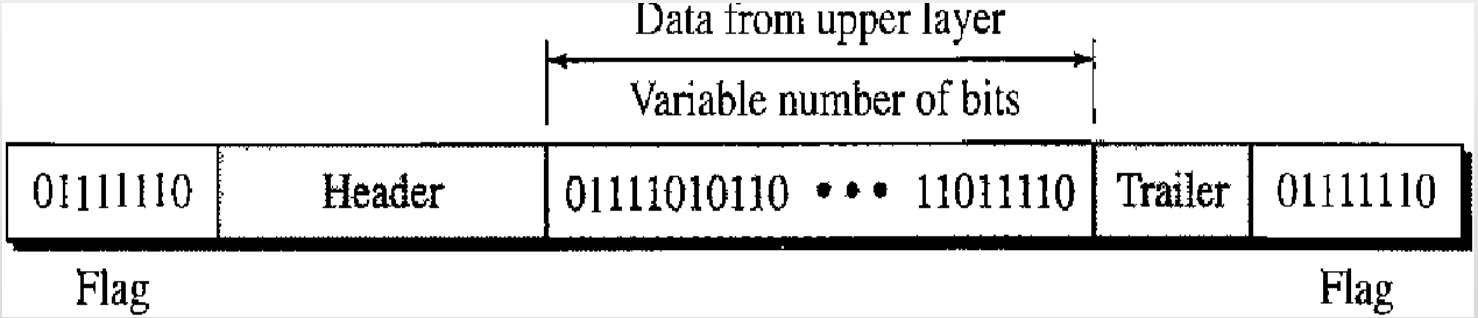
❖ Bit-Oriented Protocol:

➢ Bit-oriented protocol:-

- The data section of a frame is a sequence of bits to be interpreted by the upper layer as **text**, **graphics**, **audio**, **video**, etc.

➢ In addition to header and trailer, we need a **delimiter to separate the frame from other**.

➢ Most protocol used special 8-bit pattern **flag** 01111110 as a delimiter to define the beginning and end of frame as shown below.

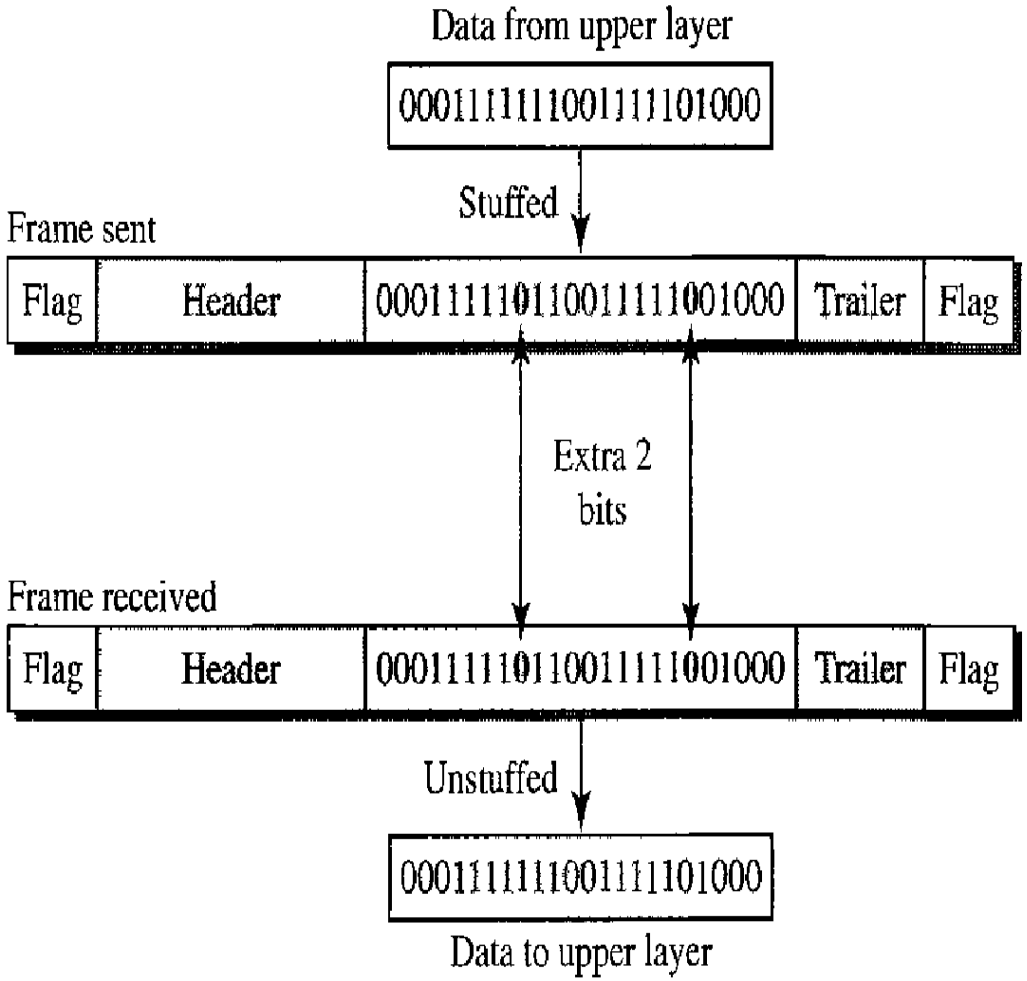




Data Link Layer: Framing: Bit-Oriented Approach

❖ Bit-Oriented Protocol: Bit-stuffing

- In bit-oriented framing, we add an extra bit called as **bit-stuffing**. Whereas in character-oriented framing we add byte-stuffing (1 extra byte).
- In bit-stuffing process, an extra bit **0** is added whenever five consecutive **1s** follow a 0 in the data, so that **receiver does not mistake a pattern** 01111110 **for a flag.**

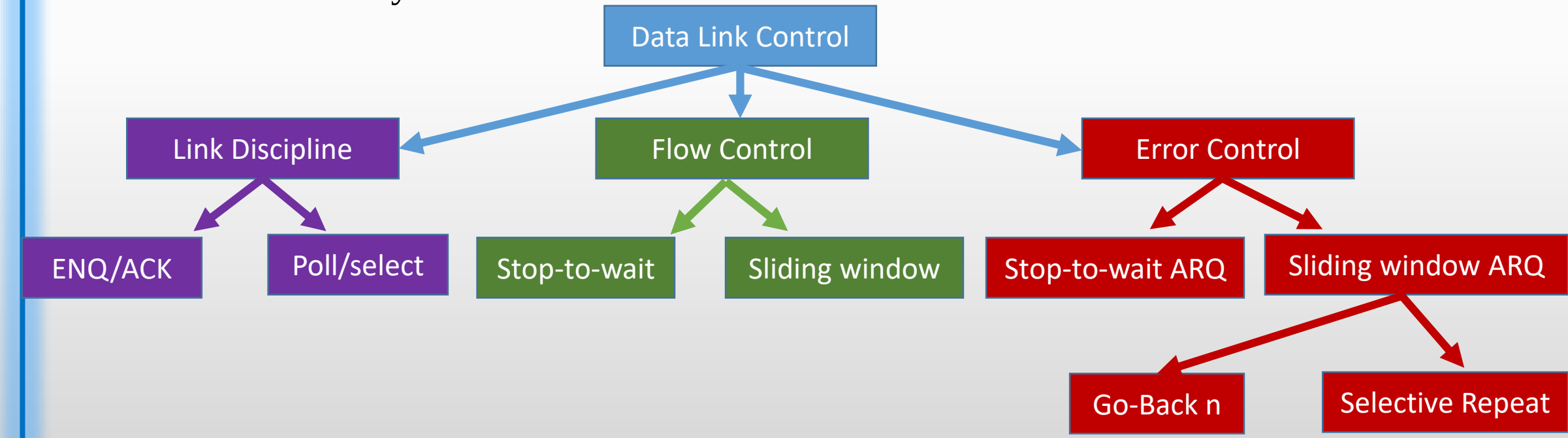




Data Link Layer: Data Link Control

❖ Data Link Control:

- ➢ The functions of data link layer are: **Line Discipline**, **Flow Control** and **Error Control**
  - ✓ Collectively these functions are known as Data Link Control





Data Link Layer: Data Link Control

Data Link Control: Line Discipline

Line discipline:

- Who should send now?

Flow Control:

- How much to be sent?

Error Control:

- How can errors be detected and corrected?



Data Link Layer: Data Link Control

Data Link Control: Line Discipline

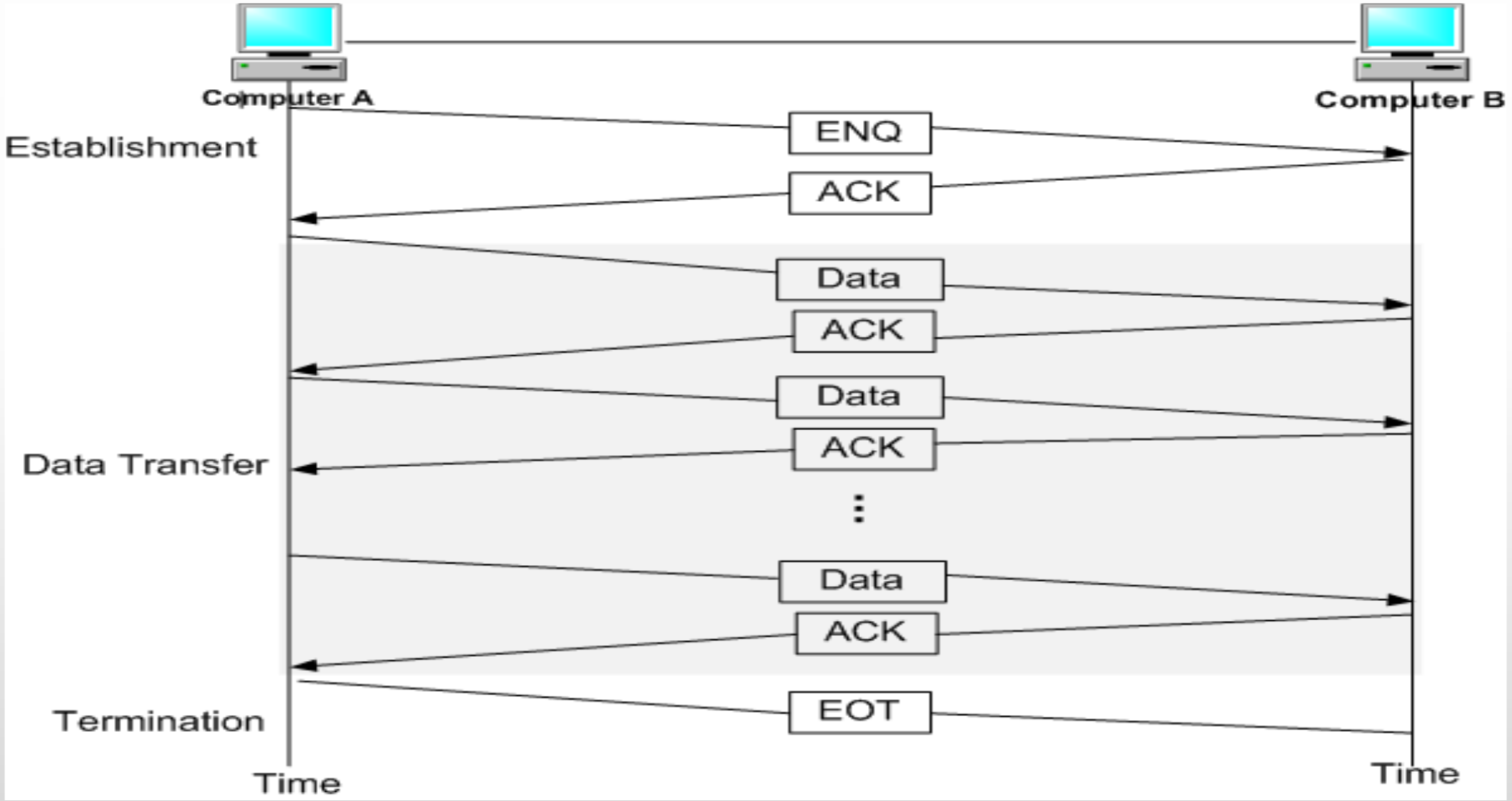
- Line discipline answer the question, **Who should send now?**
- It can be done in two ways:
  1. **Enquiry/Acknowledge**(ENQ/ACK) :– Used in peer-to-peer communication
  2. **Poll/Select** :– Used in primary-secondary communications



Data Link Layer: Data Link Control

❖ Data Link Control: Line Discipline

ENQ/ACK line Discipline





Data Link Layer: Data Link Control

❖ Data Link Control: Line Discipline → Poll/Select

- Poll/select line discipline method works with topologies where one device is designated as a **Primary Station** and the other as **Secondary Station.**
  - ✓ Multipoint configurations concept.
  - ✓ **Primary** device control the link;
  - ✓ **Secondary** device follow the instruction

- **Primary device** determine which device is allowed to use the channel at a given time.
  - ✓ Therefore, primary device is always an initiator of a session



Data Link Layer: Data Link Control

Data Link Control: Line Discipline → Poll/Select

- If Primary wants to receive data, it asks the secondary if they have any thing to send.
  - This function is called **polling.**
- If Primary wants to send data, it tells the target secondary to get ready to receive.
  - This function is called **selecting.**
- Addresses are needed in Poll/Select method.



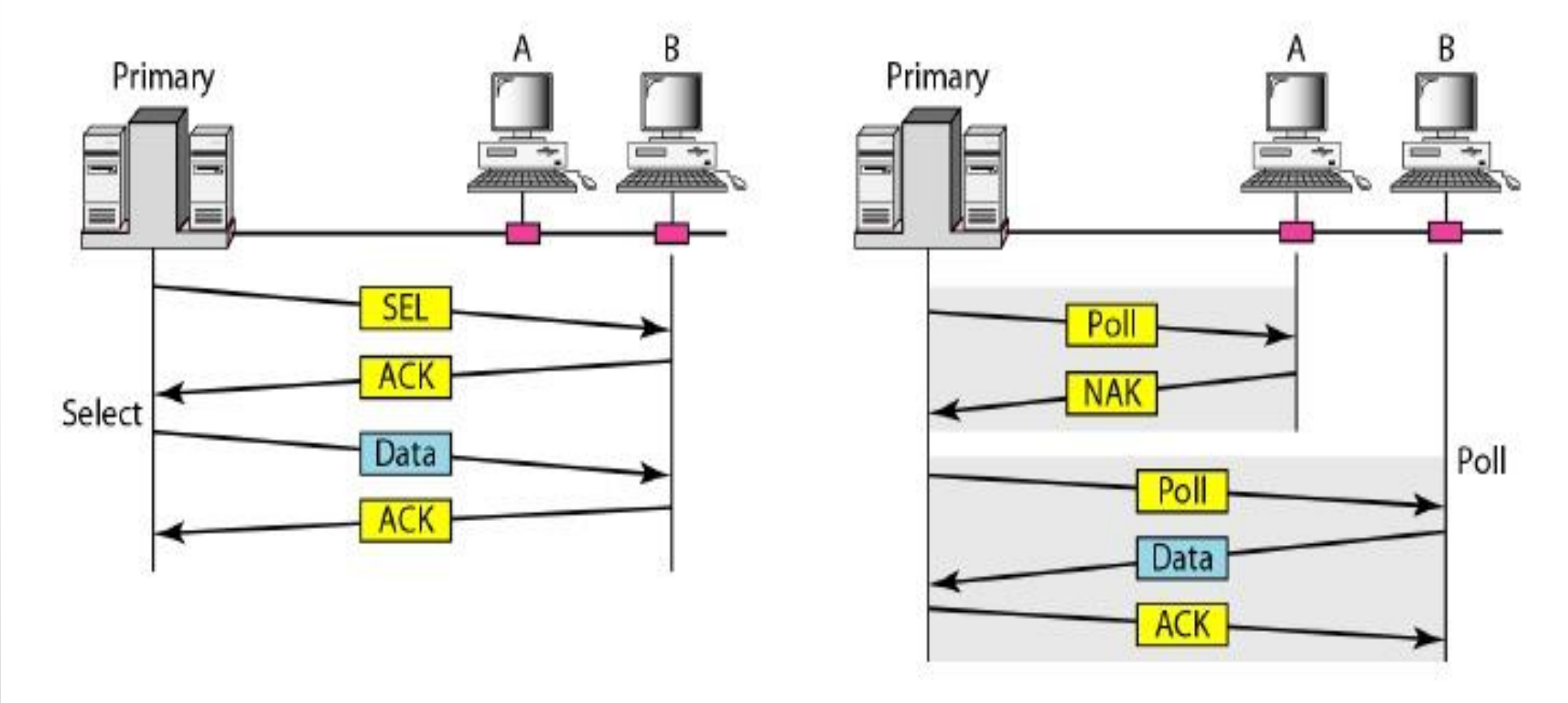
Data Link Layer: Data Link Control

Poll: Receive

- The primary device asks if secondary have got frames to send.
  - The primary initiate the session.
- Poll method follows following steps to transmit the data frame.
  1. Poll (Primary sends Poll frame to First Secondary)
  2. Data/NAK (Data frame send by first secondary device otherwise NAK frame is send)
  3. Poll (Primary sends Poll frame to next Secondary)
  4. Data/NAK and so on.

- Two possibilities for terminating the exchange: either **secondary send EOT frame** or primary says "Time's up"

Select and Poll





Data Link Layer: Data Link Control

Data Link Control: Flow Control

- Flow control mechanism is in placed in order to avoid **overwhelming receiver**.
- Flow control refers to a set of procedures used to restrict the amount of data the sender can send before waiting for acknowledgement.
- Two methods have been developed to control flow of data across communications links:
    1. **Stop-and-Wait** (Request/Reply):- send one frame at a time
    2. **Sliding window**:- send several frames at a time.



Data Link Layer: Data Link Control

❖ Data Link Control: Flow Control → Stop-and-wait

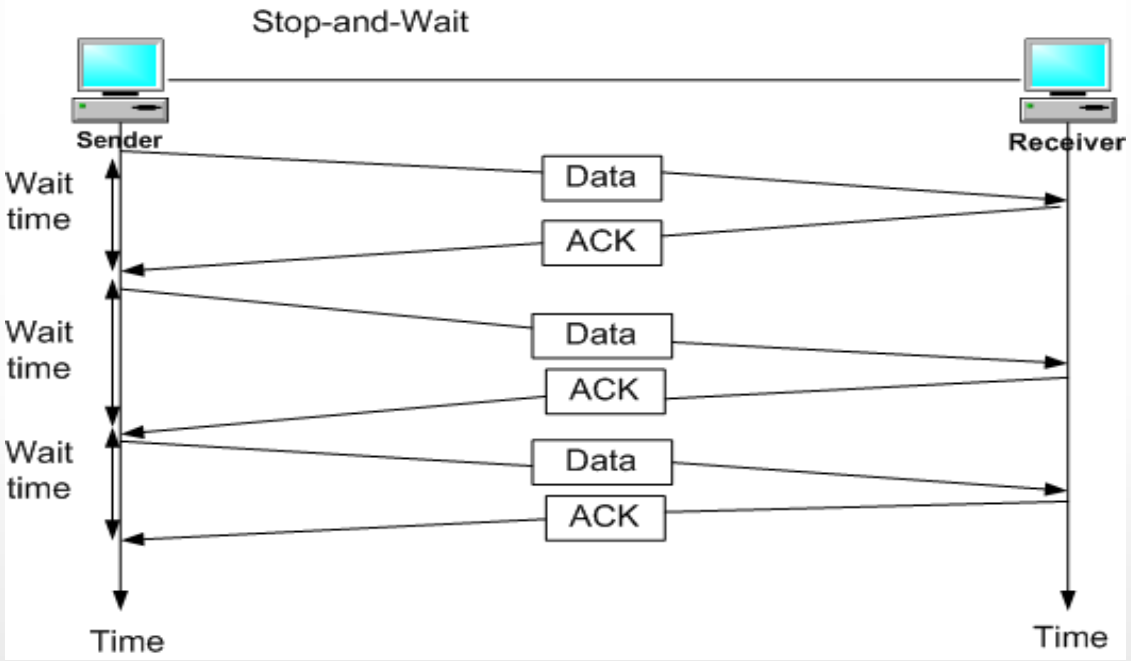
➢ Stop-and-Wait:-

- It is the simplest form of **flow control** where each frame is checked and acknowledged before the next frame is sent.
- The sender sends a frame and wait for acknowledgement from the receiver.
- After ACK frame from receiver (means willing to receive another frame), sender sends another data.



Data Link Layer: Data Link Control

❖ Data Link Control: Flow Control → Stop-and-wait



- **The disadvantage is inefficiency:-**
  - Total transmission time will be more when Data frame and ACK frames moves long distance between sender and receiver.



Data Link Layer: Data Link Control

Data Link Control: Flow Control → Sliding Window

- In the previous protocols, data frames were transmitted in one direction only.
- Therefore, in most practical situations, there is a need to transmit data in both directions.
- To improve the **link utilization**, a sliding windows protocol was developed.

- In sliding window:-
  - The sender can transmit several frames before an acknowledgement.



Data Link Layer: Data Link Control

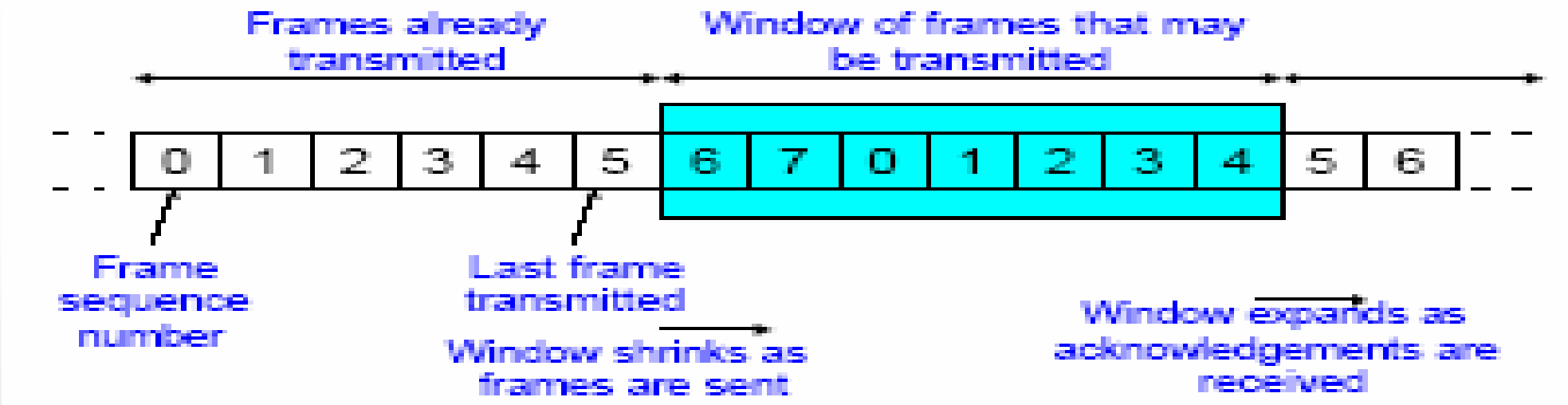
Data Link Control: Flow Control → Sliding Window

- To keep track of which frames have been transmitted and which received.
  - Sliding window introduces an identification scheme based on the size of the window.
    - ✓ The frames are numbered modulo-n, which means 0 to n-1.
    - ✓ If n=8, the frames are numbered 0,1,2,3,4,5,6,7,0,1,2,3,4,5,6,7,0,1,…..
    - ✓ The size of windows is n-1. Therefore, maximum of n-1 frames may be sent before an acknowledgement is required.

- Receiver acknowledged by sending ACK frame that include the sequence number of next frame expected.

Sender Sliding Window:

- At any instant, the sender send frames with sequence numbers in a certain range.



- Conceptually, the sender sliding window shrinks from the left when frames of data are sent and expands to the right when acknowledgments are received.



❖ Receiver Sliding Window:

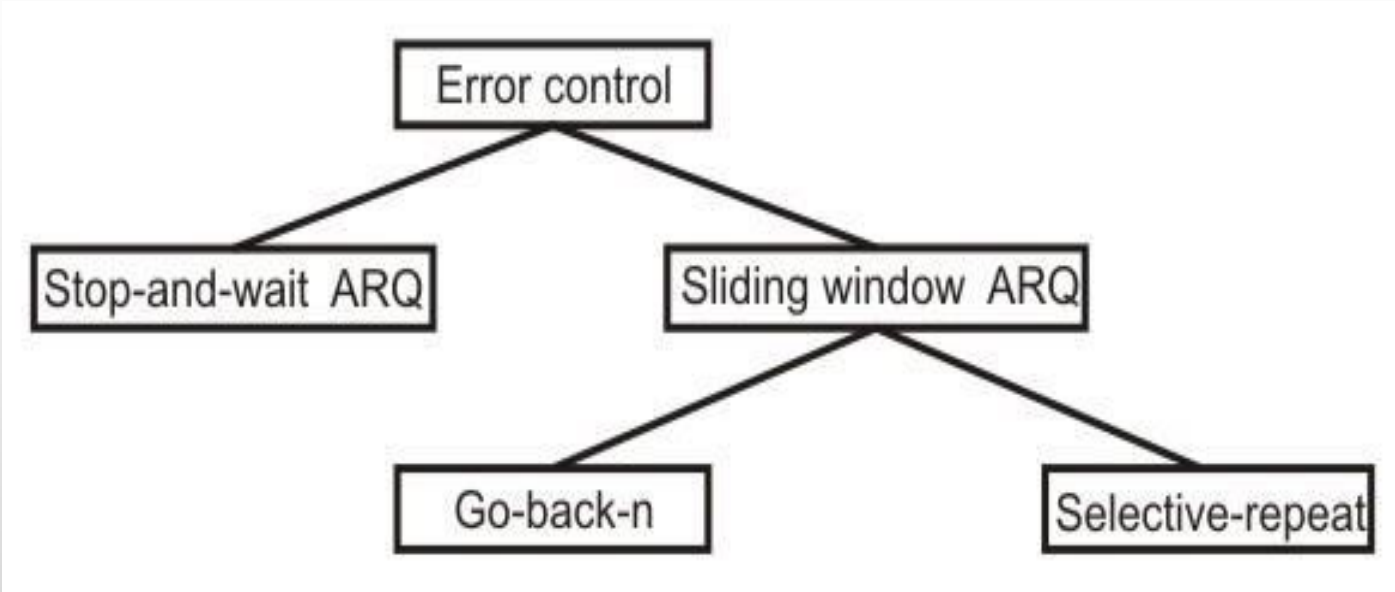
- The receiver acknowledges a frame by sending ACK frame with sequence number of the next frame expected.
- This also announces that it is prepared to received next N frames, beginning with the number specified.
- Conceptually, the receiver sliding window shrinks from the left when frames of data are received & expands to the right when acknowledgement are sent.



Data Link Layer: Data Link Control

Error Control:

- **When error is detected in a** message or packet, the *receiver sends a request to transmitter to retransmit the defected message.*
- The most popular retransmission scheme is known as **Automatic-Repeat-Request**(ARQ)





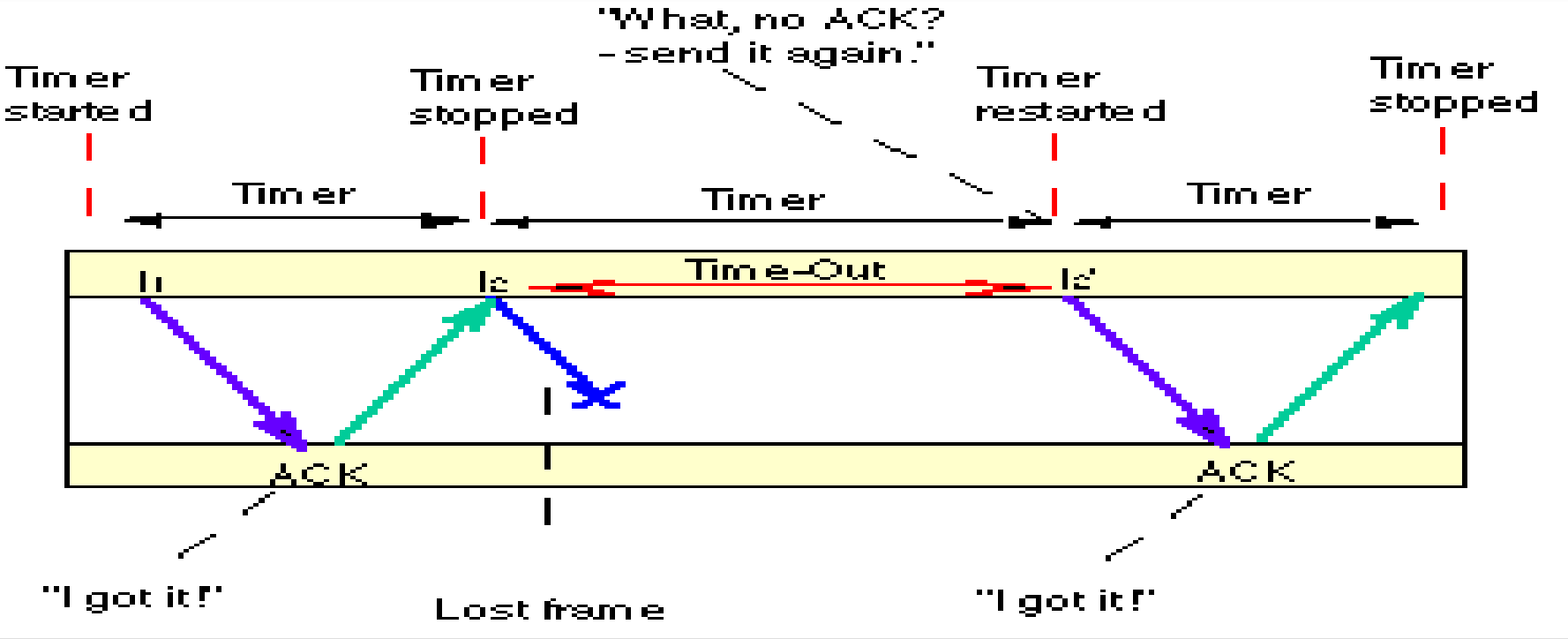
Data Link Layer: Data Link Control

Data Link Control: Error Control → Stop-and-wait ARQ

- It is a simplest among all protocols, sender transmits a frame and wait till it receives positive acknowledgment(ACK) or negative acknowledgment (NAK or NACK)
  - If ACK is receives, sender send next frame and if NAK/NACK is received it retransmit a old frame.
  - Error correction in **stop-and-wait ARQ** is done by keeping a copy of sent frame and retransmits the frame when the timer expire.

Stop-and-wait ARQ:

- Retransmission of frame due to the lost frame.





❖ Sliding window ARQ: Go-Back N ARQ

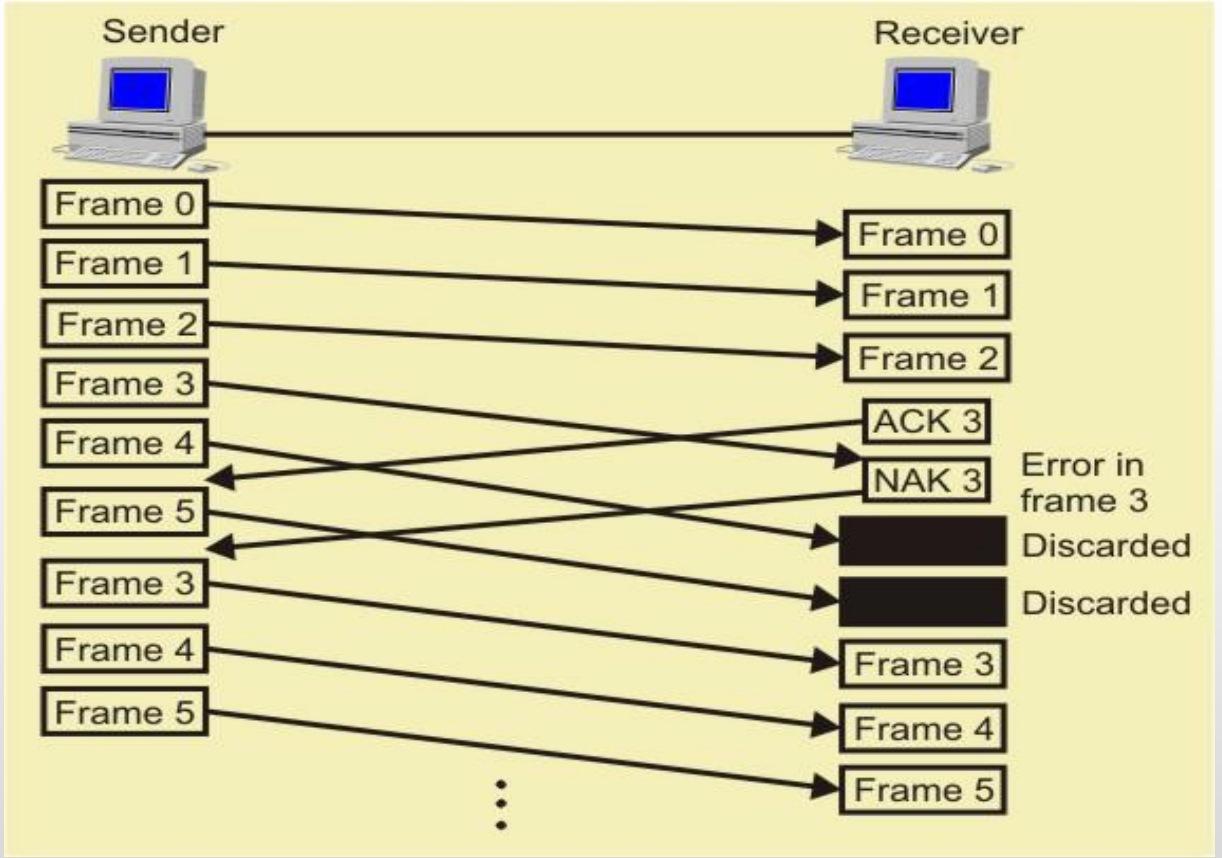
- This protocol is most popular ARQ protocol.
  - In **Go-back-N ARQ**, sender sends the frames continuously without waiting for acknowledgment. Thus sometimes it is called as **Continuous ARQ**.
- For k-bit sequence number field, 2^k-1 frames can be send without receiving acknowledgment.
  - Therefore, maximum window size = 2^k-1
- For example, 3-bit field sequence number will be 0,1,2,3,4,5,6,7,0,1,2,3,…. and window size=8.



Data Link Layer: Data Link Control

❖ Sliding window ARQ: Go-Back N ARQ

- ➢ The frames in error Go-Back N ARQ.

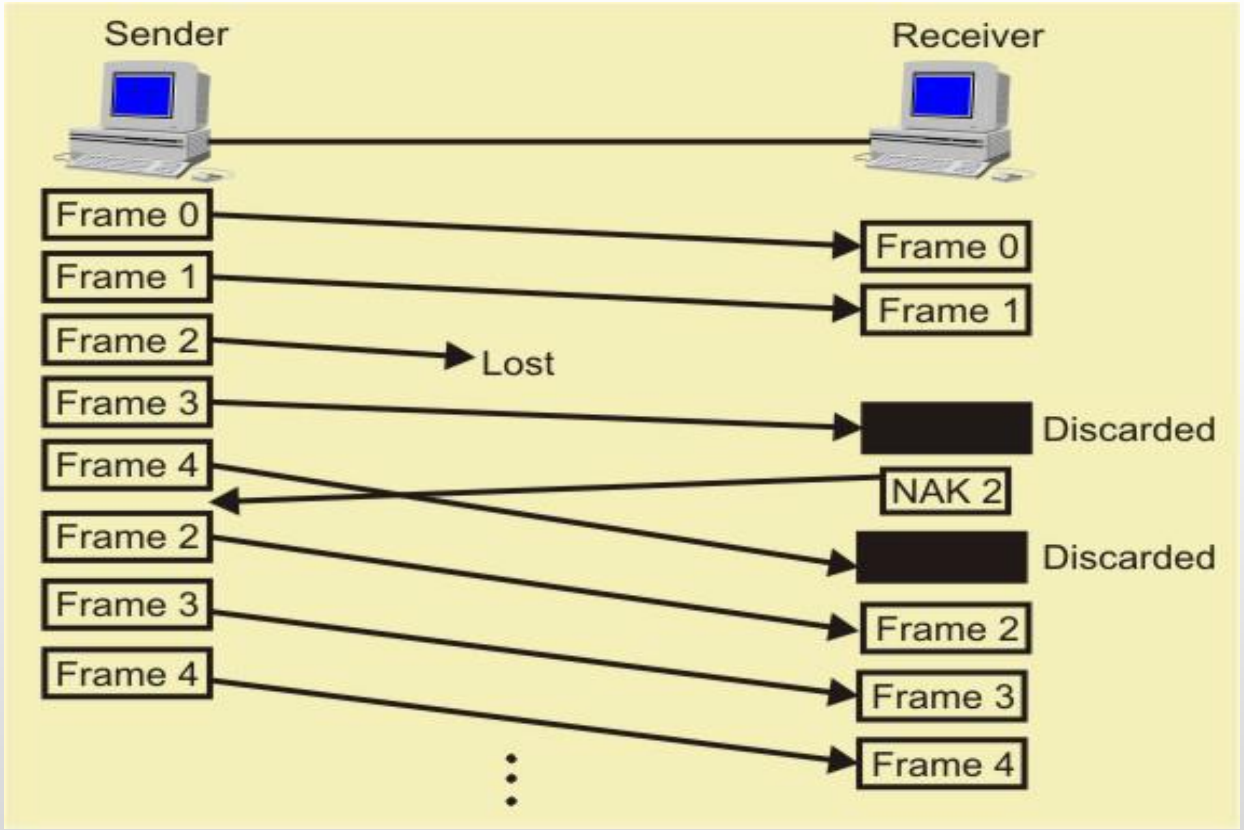




Data Link Layer: Data Link Control

❖ Sliding window ARQ: Go-Back N ARQ

- The lost frames in error Go-Back N ARQ.

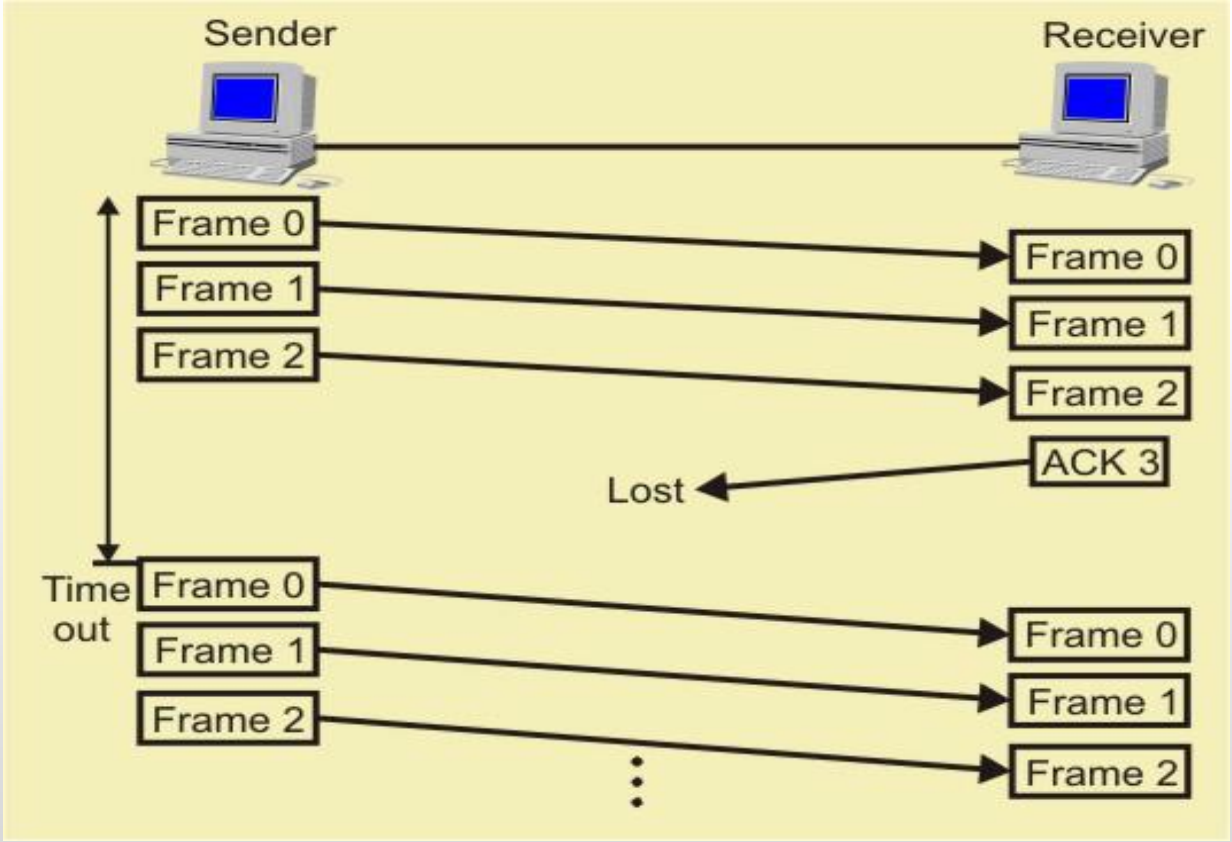




Data Link Layer: Data Link Control

❖ Sliding window ARQ: Go-Back N ARQ

- The lost ACK in error Go-Back N ARQ.





❖ Sliding window ARQ: Selective Repeat ARQ

- Selective Repeat ARQ error control mechanism does not resent N frames when just one frame is damaged.
  - ✓ Only damaged frame is resent (but **Go-back-N ARQ resends the N frames when one frame is damaged**.)
- Most efficient among the ARQ schemes..



Recapitulation

- **We have learned:**
  - Frame and framing
  - Character-oriented and Bit-oriented protocol in framing.
  - Bit stuffing and unstuffing
  - Data link control – **Line Discipline, Flow Control and Error Control.**
  - Types of each data link control.



References

References:

1. Stallings, W., Data & Computer Communication, Prentice-Hall
2. Tanenbaum, Computers Networks, PHI
3. Forouzan, B.A., Data Communication & Networkings, 4th Edition.
4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan



See you in next class.